Data Structures Lab 14: Hash Tables

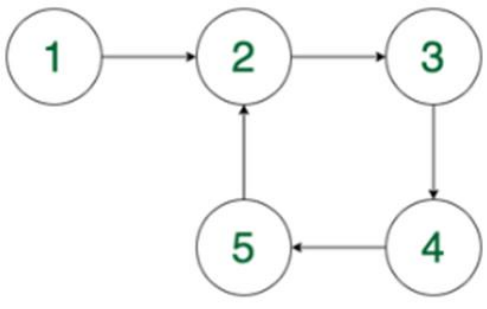
Question 1: Write a C++ code to implement ‘Chaining” in a Hash Tabel. You will need to make the following classes:

1. Class Node {
   int key
   Int value
   Node *next;
   }
   You will need to make the constructor of this class as well.

2. Class HashTable {
   Int num of elements
   Int capacity
   Node ** arr_of_linkedlist
   }
3. Int Hash_Function(int key) // You can use the division hash function
4. Void insert (int key , int value)
5. Void remove (int key)
6. Int search()
7. ~HashTable()

Question 2:

Given a linked list, check if the linked list has a loop (cycle) or not. The below diagram shows a linked list with a loop.



Linked List with Loop

The idea is to insert the nodes in the hashtable and whenever a node is encountered that is already present in the hashtable then return true.